

MAC 233: External LED Lab

Dr Dan Brennan

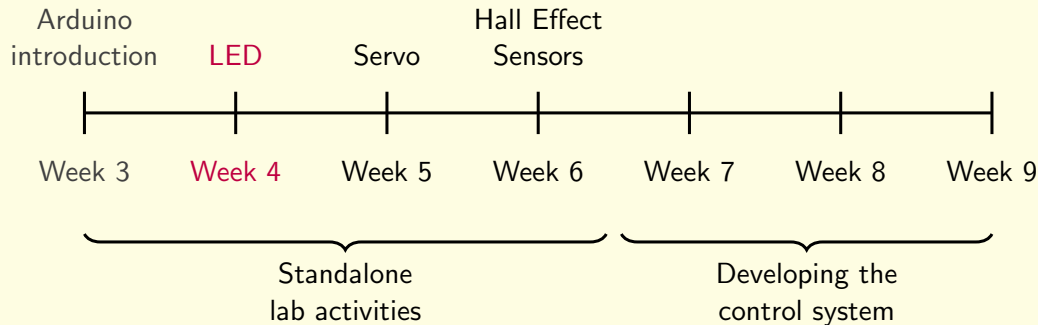
School of Mechanical, Aerospace, and Civil Engineering,
University of Sheffield, UK

MAC 233: **22/10/2025**



University of
Sheffield

MAC 233: LABS OVERVIEW



VARIABLES: SCOPE

```
// not inside any functions: Global scope  
// accessible from anywhere  
int wheel_rotations = 0;  
  
void setup() {  
    // inside the setup function: Setup scope  
    // only accessible from setup function  
    int setup_rotations = 1;  
}
```

TODAY'S LAB: PREREQUISITES

This worksheet builds upon the Lab 1 worksheet. You will have needed to complete the Lab 1 worksheet before you can continue with this lab.

Out of your Arduino pack, you will need: 1 Arduino, 1 breadboard, 1 LED, 1 220Ω resistor, and 1 jumper cable.

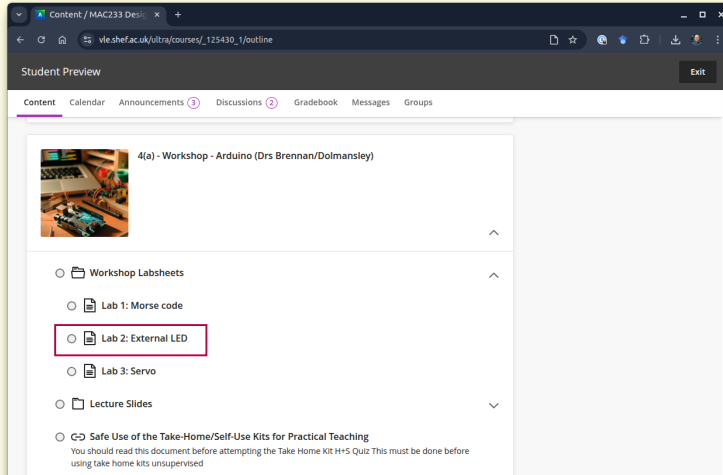
TODAY'S LAB: OBJECTIVES

The objective of today's lab is to learn how to use an electronic breadboard to create a circuit and to control an external output on the Arduino.

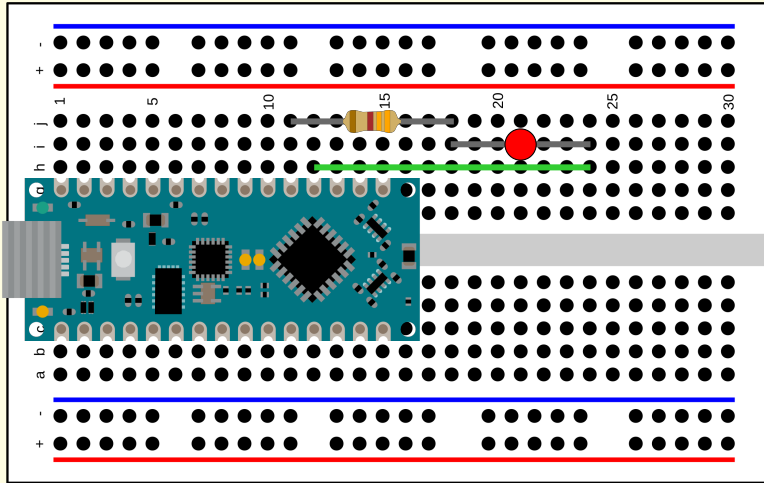
We will be taking the scenario from the previous worksheet and changing it to communicate with an external LED.

LAB 2: EXTERNAL LED

Login to Blackboard: <https://vle.shef.ac.uk/>



WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>, Resistor source <https://freesvg.org/resistor-330-ohm>.